



RETRIEVAL-AUGMENTED GENERATION

73.69 - LARGE LANGUAGE MODELS - 2026

Ing. Eugenia Piñeiro, Ing. Marina Fuster

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01

INTRODUCCIÓN

Desafíos de los LLMs

INFORMACIÓN DESACTUALIZADA



Búsqueda realizada en 3 sitios

Sam Altman, the CEO of OpenAI, was sudden decision by OpenAI's board w as well as concerns about the directic style, which emphasized commerciali mission to ensure AI development remaini safe and beneficial to humanity (Data Science Deia) (UNSW Business School: BusinessThir



As an AI developed by OpenAI, my tra can retrieve up-to-date information u regarding Sam Altman's dismissal, wh (UNSW Business School: BusinessThir information, feel free to ask!



Búsqueda realizada en 3 sitios



"Sam Altman dismissal date"
bing.com



Yahoo News - Latest News & Headlines — A timeline of Sam Altman's firing from OpenAI – and the...
news.yahoo.com



UNSW Business School: BusinessThink — Who is OpenAI's Sam Altman, and why does it matter that ...
businessthink.unsw.edu.au

please dont use yahoo as a source

< 2/2 >



Búsqueda realizada en 4 sitios



"Sam Altman dismissal date November 2023 OpenAI news"
bing.com



Wikipedia — Removal of Sam Altman from OpenAI - Wikipedia
en.wikipedia.org



OpenAI — Review completed & Altman, Brockman to continue to lead OpenAI | OpenAI
openai.com



Engadget — OpenAI fires CEO Sam Altman as 'board no longer has confidence' in his leadership
engadget.com

INFORMACIÓN FALSA

summarize Yann Lecun's most recent paper "LLMs are not the future" and give me the paper citation please



Yann LeCun's paper titled "LLMs are Not the Future" is a critical examination of large language models (LLMs) and their limitations. In this paper, LeCun argues that while LLMs have made significant advancements in natural language processing, they are not the ultimate solution for artificial general intelligence (AGI). He highlights several issues with current LLMs, including their reliance on vast amounts of data, their inability to understand context deeply, and their lack of true reasoning capabilities.

LeCun suggests that future progress in AI will require a shift towards more robust models that integrate both learning from data and structured reasoning. He advocates for a hybrid approach combining symbolic AI with neural network-based methods to address these shortcomings and move closer to AGI.

Here's the citation for the paper:

LeCun, Y. (2024). "LLMs are Not the Future." Journal of Artificial Intelligence Research.

Please verify the exact publication details from a reliable source as they may vary.



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PROJECTS



Yann LeCun

Yann LeCun

Chief AI Scientist, Meta
Jacob T. Schwartz Professor of Computer Science, Data Science, Neural Science, and Electrical and Computer Engineering, [New York University](#).
ACM Turing Award Laureate, (sounds like I'm bragging, but a condition of accepting the award is to write this next to my name)
Member, National Academy of Engineering, National Academy of Sciences, Académie des Sciences
Fellow, ACM, AAAI, AAAS, SIF

last updated: 2024-07-14

Social Networks

Facebook: [yann.lecun](#)
Twitter: [@ylecun](#)
LinkedIn: [yann-lecun](#)
Threads: [@yannlecun](#)



Yann LeCun

VP & Chief AI Scientist at Meta

Ver perfil completo



Yann LeCun • Siguiendo
VP & Chief AI Scientist at Meta
1 mes •

Nice article in Financial Times where I explain that Auto-Regressive LLM are insufficient to reach human-level intelligence (or even cat-level intelligence).

But alternative architectures that I call "objective driven" may reach human-level intelligence one day.

They use world models based on JEPa (Joint Embedding Predictive Architectures, which are not generative).

With this, we may have systems that

1. understand the physical world
2. have persistent memory
3. can reason
4. can plan, perhaps hierarchically.

Four essential characteristics necessary for intelligent behavior, which humans and many animals exhibit.

INFORMACIÓN FALSA

give me Yann Lecun's most recent paper "LLMs don't work" and reference the paper

Búsqueda realizada en 5 sitios

Yann LeCun's recent paper, titled "Augmented Language Models: limitations and potential improvements for Large Language Models", provides an in-depth analysis of how LLMs can be augmented with reasoning capabilities to perform more effectively. It highlights the need for LLMs to go beyond simple pattern matching and incorporate mechanisms for better problem decomposition and task planning.

You can access the full paper [here \(arXiv\)](#) ([OpenReview](#)).

A Path Towards Autonomous Machine Intelligence Version 0.9.2, 2022-06-27

Yann LeCun
Courant Institute of Mathematical Sciences, New York University yann@cs.nyu.edu
Meta - Fundamental AI Research yann@fb.com

June 27, 2022

Abstract

How could machines learn as efficiently as humans and animals? How could machines learn to reason and plan? How could machines learn representations of percepts and action plans at multiple levels of abstraction, enabling them to reason, predict, and plan at multiple time horizons? This position paper proposes an architecture and training paradigms with which to construct autonomous intelligent agents. It combines concepts such as configurable predictive world model, behavior driven through intrinsic motivation, and hierarchical joint embedding architectures trained with self-supervised learning.

Keywords: Artificial Intelligence, Machine Common Sense, Cognitive Architecture, Deep Learning, Self-Supervised Learning, Energy-Based Model, World Models, Joint Embedding Architecture, Intrinsic Motivation.

INFORMACIÓN NO PERSONALIZADA

BASE MODEL

What was the name of that movie Chris emailed me about last year?



Sorry, I don't have access to any specific information about emails or ...

BASE MODEL + USER DATASET

What was the name of that movie Chris emailed me about last year?



User dataset



Sentence xformer



Vector Library

Chris Thomson emailed you about "The Fall" on October 23rd last year. The Fall is an adventure fantasy film released in 2006, starring...

LIMITACIONES

- 1) Capacidad para comprender el *mundo físico*
- 2) Capacidad de *memorización*
- 3) Habilidades de *razonamiento*
- 4) Capacidad de *planificación*



[Yann Lecun Podcast](#)

“THE CURSE OF THE LLMs”



SOLUCIÓN

Combinar:

Memoria paramétrica

Información que se encuentra en los parámetros del modelo durante en el entrenamiento

Memoria no-paramétrica

Estructura de datos externa que agrega información al modelo en el momento de la inferencia

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis^{†‡}, Ethan Perez^{*},

Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

[†]Facebook AI Research; [‡]University College London; ^{*}New York University;
plewis@fb.com

Abstract

Large pre-trained language models have been shown to store factual knowledge in their parameters, and achieve state-of-the-art results when fine-tuned on downstream NLP tasks. However, their ability to access and precisely manipulate knowledge is still limited, and hence on knowledge-intensive tasks, their performance lags behind task-specific architectures. Additionally, providing provenance for their decisions and updating their world knowledge remain open research problems. Pre-trained models with a differentiable access mechanism to explicit non-parametric memory have so far been only investigated for extractive downstream tasks. We explore a general-purpose fine-tuning recipe for retrieval-augmented generation (RAG) — models which combine pre-trained parametric and non-parametric memory for language generation. We introduce RAG models where the parametric memory is a pre-trained seq2seq model and the non-parametric memory is a dense vector index of Wikipedia, accessed with a pre-trained neural retriever. We compare two RAG formulations, one which conditions on the same retrieved passages

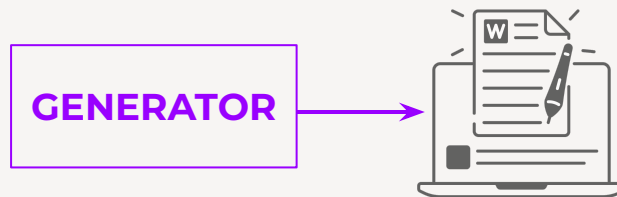
RETRIEVAL-AUGMENTED GENERATION (RAG)

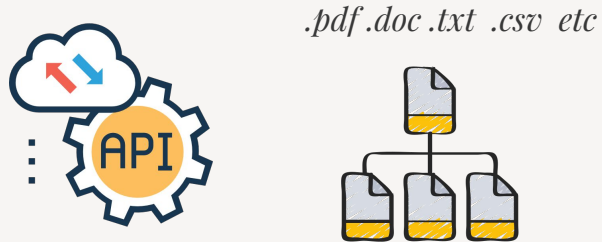
Modelo de arquitectura para mejorar la eficacia de los LLMs

Retriever: obtener el contexto relevante de una extensa base de conocimiento externa (docs collection)



Generator: actuando sobre la información recuperada, crea la respuesta final agregando el *contexto* adicional.



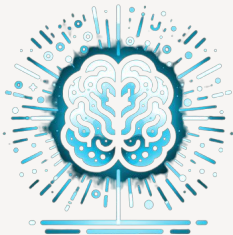


BENEFICIOS DE RAG

Conocimiento expandido

- Acceso a múltiples fuentes
- Fácil actualización
- Menos costoso que re-entrenar

< hallucinations



GENERATOR

Source cites



context awareness

BENEFICIOS DE RAG

Aumento de confianza

- Transparencia: citas de fuentes
- Contexto: respuestas precisas
- Reduce alucinaciones

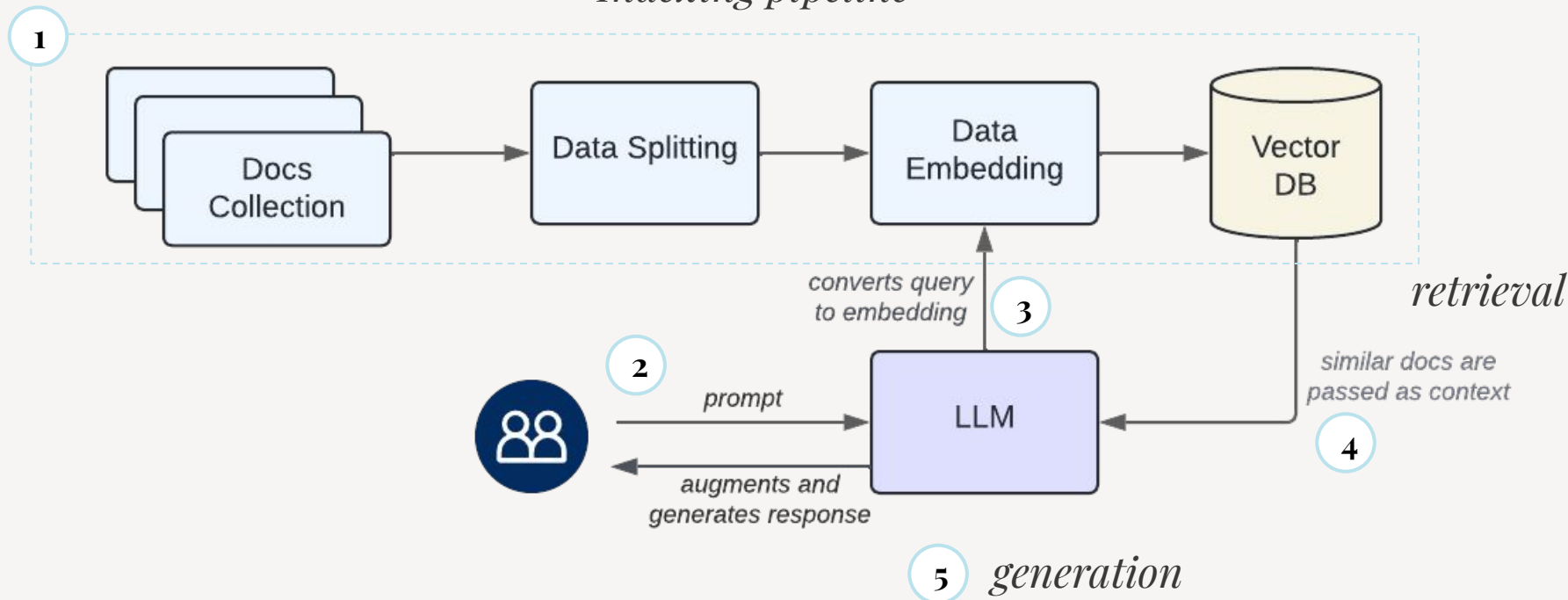
02

ARQUITECTURA

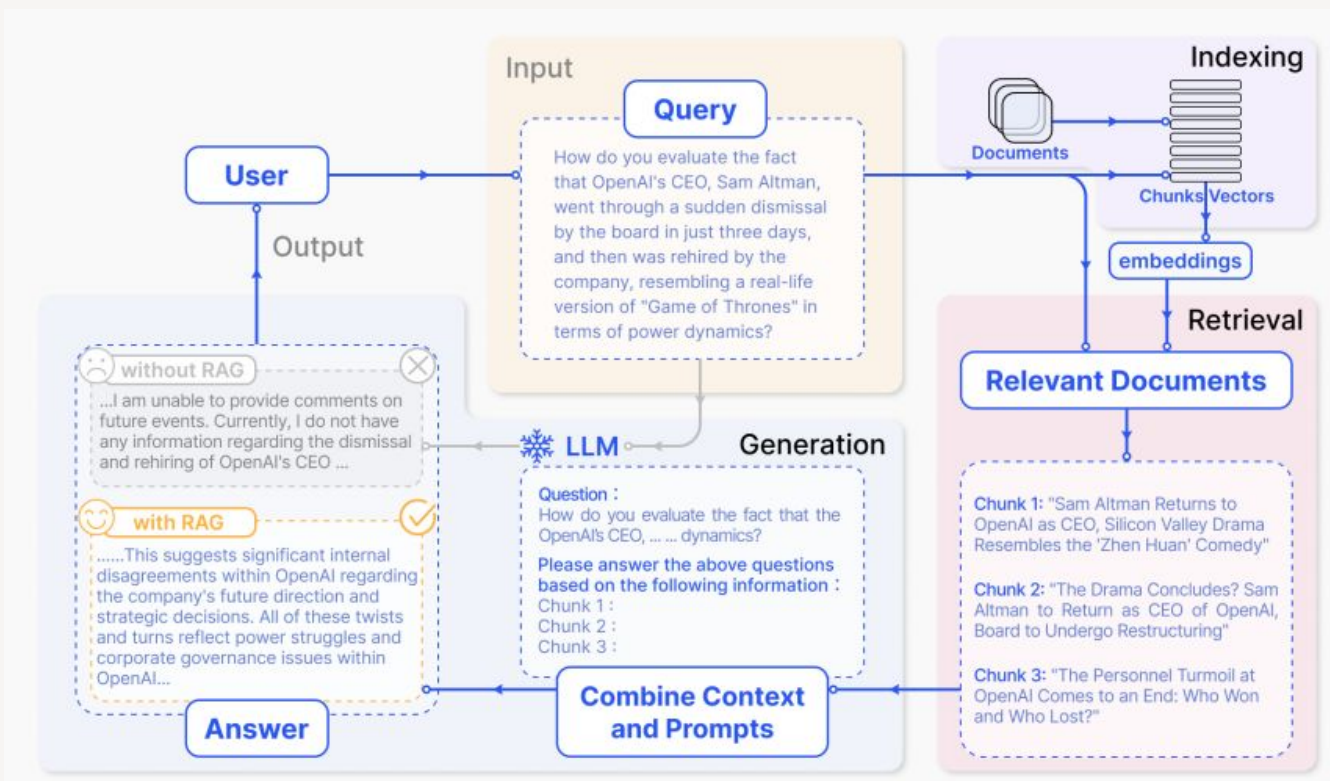
RAG Pipeline

PIPELINE

Indexing pipeline



EJEMPLO



02.1

INDEXING PIPELINE

Data Loading, Text Splitting

INDEXING PIPELINE

01

LOADER

02

SPLITTER

03

EMBEDDING
ALGORITHM*

04

VECTOR
STORE

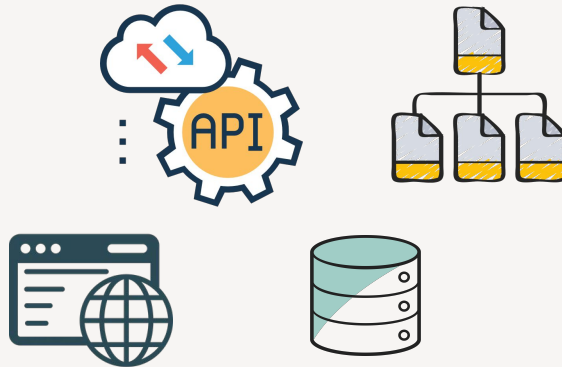
**NOTA: Para algoritmos de embedding dirigirse a clase 2*

DATA LOADING

El refuerzo recae en construir una **buena base de conocimiento**.

Obtener información de múltiples fuentes:

- APIs
- Web scraping
- Data en formato .json .csv
- File systems (.pdf .doc .txt etc)
- Bases de datos
- ... entre otros



No es un proceso trivial (**data cleaning**, transformaciones, parsing, etc)

TEXT SPLITTING

Dividir cada documento en “*chunks*” (partes más pequeñas)

BÚSQUEDA

Mejora la indexación

Búsqueda de
documentos eficiente

CONTEXTO

LLMs tienen cantidad
limitada de tokens

“context window”

TEXT SPLITTERS

CHARACTER SPLITTER

- El texto se divide por un caracter predefinido Ej: \n\n
- La longitud de chunk se mide por: cantidad de caracteres

chunk 1

El Paper 'Attention is All You Need' [6] publicado en el año 2017 introduce una arquitectura de Encoders y Decoders basada en un mecanismo de atención, a la que denominaron 'Transformer'. *Un Transformer es una red neuronal cuyo principal objetivo es transformar una secuencia en otra.*

A diferencia de otros tipos redes], permite ejecutar tareas de forma paralela reduciendo el tiempo requerido para entrenar. A su vez, puede recordar viejas conexiones en secuencias de gran tamaño evitando el problema del gradiente que va desapareciendo o crece demasiado. (vanishing/exploding gradients).[2]

chunk 2

TEXT SPLITTERS

OVERLAP

- Tokens que se superponen entre chunks
- Mantener la continuidad contextual entre los chunks.
- Porcentaje del “*chunk_size*”

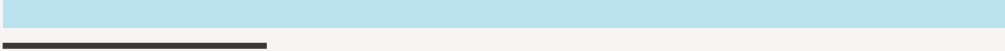
chunk 1

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transformar una secuencia en otra. A diferencia de otros tipos de redes, permite ejecutar tareas de forma paralela reduciendo el tiempo requerido para entrenar. A su vez, puede recordar viejas conexiones en secuencias de gran tamaño evitando el problema del gradiente que va desapareciendo.

chunk 2

¿Qué problema puede provocar
dividir por caracteres?



TEXT SPLITTERS

RECURSIVE CHARACTER TEXT SPLITTER

- Intenta preservar mejor la relación semántica
- Divide el texto recursivamente por múltiples caracteres respetando una jerarquía (párrafos, oraciones, palabras)

[‘\n\n’, ‘\n’ , ‘.’ , ‘ ‘]

Cuidado! Potencial “oversplitting”

TEXT SPLITTERS

RECURSIVE CHARACTER TEXT SPLITTER

```
separators = [ '\n\n', '\n', '.', ' ' ]
```

Recursivamente por cada *separator*

1. Intenta dividir por *separator*

```
    if(lo encuentra y len(chunk) > CHUNK_SIZE)  
        ese chunk lo intenta dividir por separator+1
```

TEXT SPLITTERS

RECURSIVE CHARACTER TEXT SPLITTER

```
separators = [ '\n\n', '\n', '.', ' ' ]
```

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TEXT SPLITTERS

RECURSIVE CHARACTER TEXT SPLITTER

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```

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\n\n

A diferencia de otros tipos redes, permite ejecutar tareas de forma paralela reduciendo el tiempo requerido para entrenar.
\nA su vez, puede recordar viejas conexiones en secuencias de gran tamaño evitando el problema del gradiente que va desapareciendo.

TEXT SPLITTERS

RECURSIVE CHARACTER TEXT SPLITTER

separators = ['\n\n', '\n' , ' .' , ' ']

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*A diferencia de otros tipos redes, permite ejecutar tareas de forma paralela reduciendo el tiempo requerido para entrenar.***\n**

A su vez, puede recordar viejas conexiones en secuencias de gran tamaño evitando el problema del gradiente que va desapareciendo.

TEXT SPLITTERS

RECURSIVE CHARACTER TEXT SPLITTER

separators = ['\n\n', '\n', '.', ' ']

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Un Transformer es una red neuronal cuyo principal objetivo es *transformar una secuencia en otra.*

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A su vez, puede recordar viejas conexiones en secuencias de gran tamaño evitando el problema del gradiente que va desapareciendo.

TEXT SPLITTERS

TOKEN SPLITTER

- *TikToken (BPE tokenizer):*
 - Divide el texto según un caracter pero la longitud la determina la cantidad de tokens

CONTEXT SPLITTER

- Preserva el contexto y la estructura del documento
- HTML, Markdown, Latex, [code](#)

TEXT SPLITTING - CONSIDERACIONES

**¿Cuál es la naturaleza del contenido
que estamos dividiendo?**

Si el split es random → pierdo contexto → incoherencia

TEXT SPLITTING - CONSIDERACIONES

¿Cuál es context window del modelo?

Muy pequeño

Si *“chunk_size”* <<< *“context_window”*

No tiene en cuenta las relaciones entre sentencias

Muy grande

Si *“chunk_size”* > *“context_window”* → truncate

→ respuestas incompletas o ruido, difícil encontrar una respuesta precisa, búsqueda lenta.

TEXT SPLITTING - CONSIDERACIONES

¿Qué se espera de la complejidad del prompt?

¿El usuario final va a ser preguntar largas y complejas o simples y cortas?

Queremos maximizar la similitud entre

- embedded user query
- embedded chunk

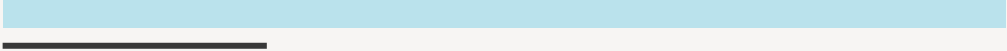
TEXT SPLITTING - CONSIDERACIONES

Trade-off

Performance vs accuracy

Experimentar y evaluar

¿Qué ventajas y desventajas
tiene el *text splitting*?

A decorative graphic consisting of two horizontal lines. The top line is light blue and the bottom line is dark blue, both spanning the width of the text above them.

TEXT SPLITTING

VENTAJAS

- Mejora la eficiencia del retriever (focus similarity search)
- Flexibilidad en el retriever (al agarrar distintos chunks la respuesta es más específica)

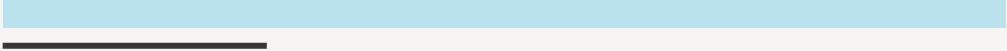
DESVENTAJAS

- Agrega costo computacional
- Elegir el tamaño de chunk y overlap es experimental

02.2

RETRIEVAL

¿Qué se debe tener en cuenta
para hacer un *efficient retrieval*?



RETRIEVAL

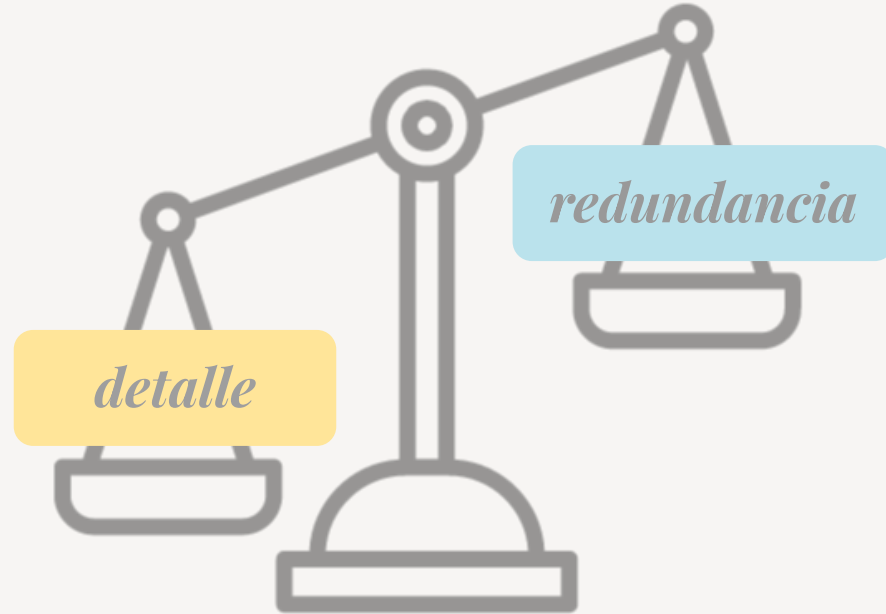
SOURCE

RAG depende de data externa

- *Structured Data*
texto, cross-lingual text, domain-specific
- *Semi-Structured data*
texto y tablas (pdf) [TableGPT](#)
- *Unstructured data*
web dump, knowledge graphs. [KnowledgeGTP](#) y [GraphGPT](#)

RETRIEVAL

GRANULARITY



03

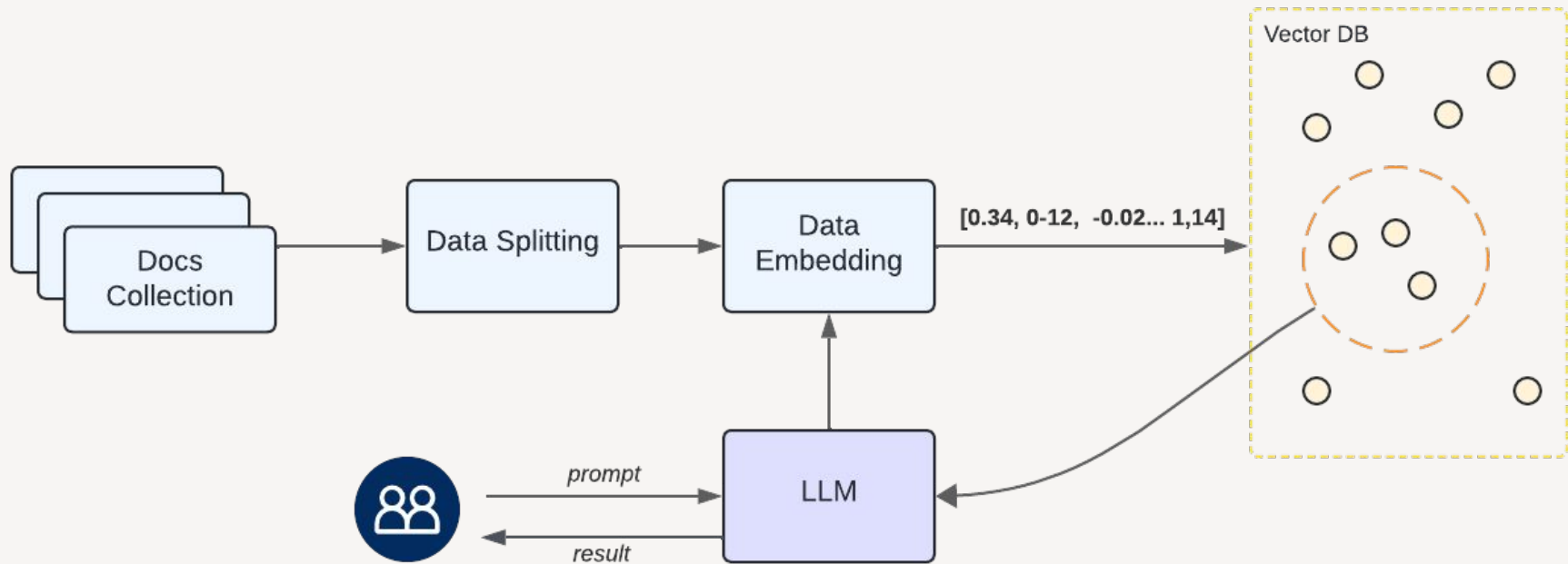
VECTOR STORE

Bases de Datos Vectoriales

BASE DE DATOS VECTORIAL

Capacidad de **almacenar** y recuperar **vectores** como puntos de alta dimensión

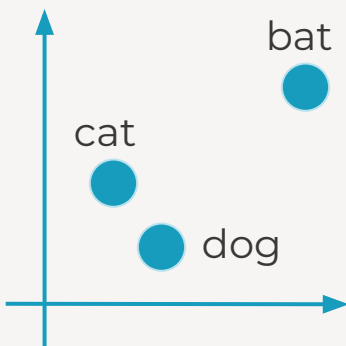
Búsqueda de similitud eficiente y rápida en espacio N-dimensional



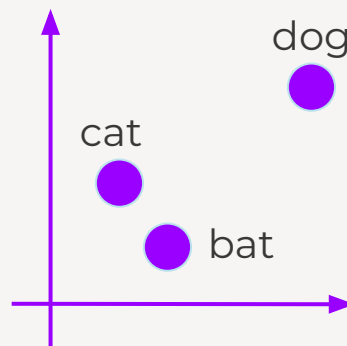
SIMILITUD

Query: cat

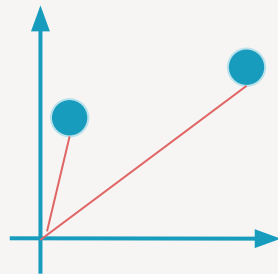
Embedding: [1,2]



id	animal	embedding
1	dog	[2, 1]
2	bat	[0, 1]



SIMILITUD



Sean dos vectores $\mathbf{a}$, $\mathbf{b}$:

Distancia Euclídea

$$d(\mathbf{a}, \mathbf{b}) = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$$

Producto Escalar

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i$$

Distancia Coseno

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{||\mathbf{a}|| \cdot ||\mathbf{b}||}$$

SIMILITUD

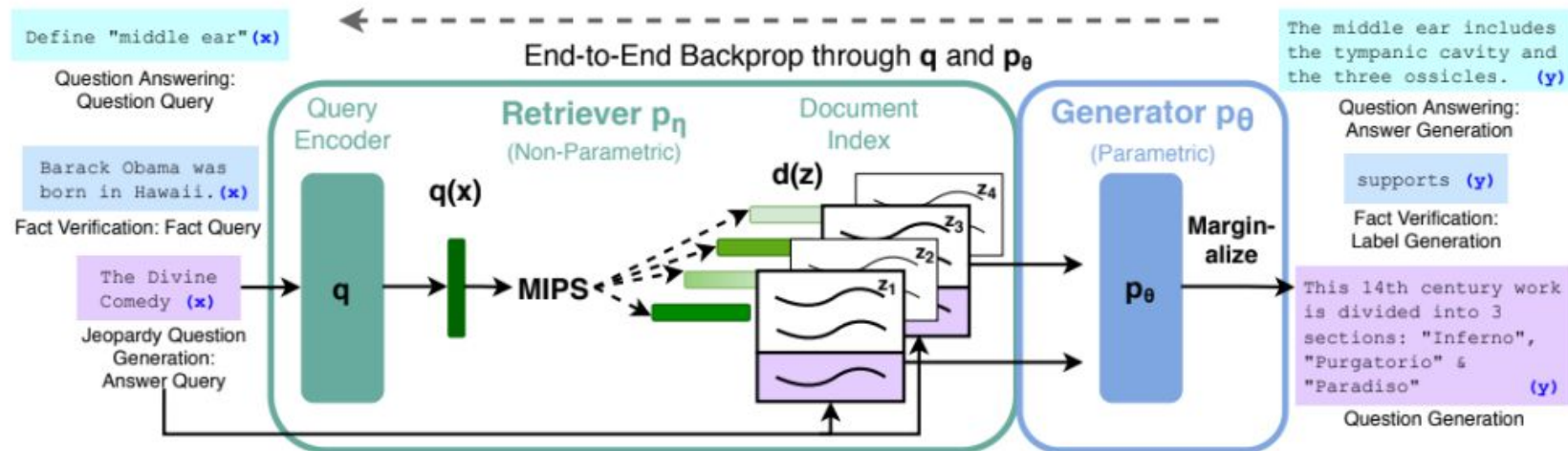


Figure 1: Overview of our approach. We combine a pre-trained retriever (*Query Encoder* + *Document Index*) with a pre-trained seq2seq model (*Generator*) and fine-tune end-to-end. For query x , we use Maximum Inner Product Search (MIPS) to find the top-K documents z_i . For final prediction y , we treat z as a latent variable and marginalize over seq2seq predictions given different documents.

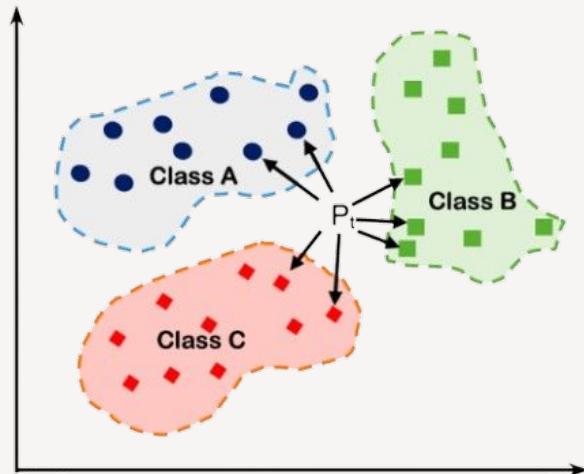
APPROXIMATE NEAREST NEIGHBOUR

K-NN → Algoritmo de Machine Learning que busca los K “vecinos” más cercanos según una medida de similitud

ANN → Encuentra un punto muy cercano pero no necesariamente el más cercano

Trade-off: velocidad vs precisión

K Nearest Neighbors



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Build knowledgeable AI

Pinecone serverless lets you deliver remarkable GenAI applications faster, at up to 50x lower cost.

[Get Started](#)[Contact Sales](#)

CHROMADB



14k



838 online

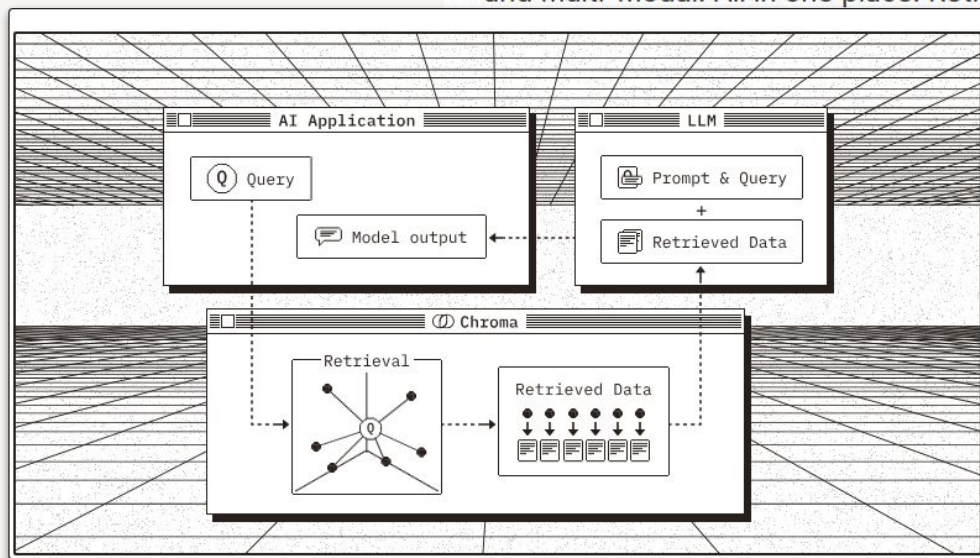
Docs Github Community Labs Careers

Get started

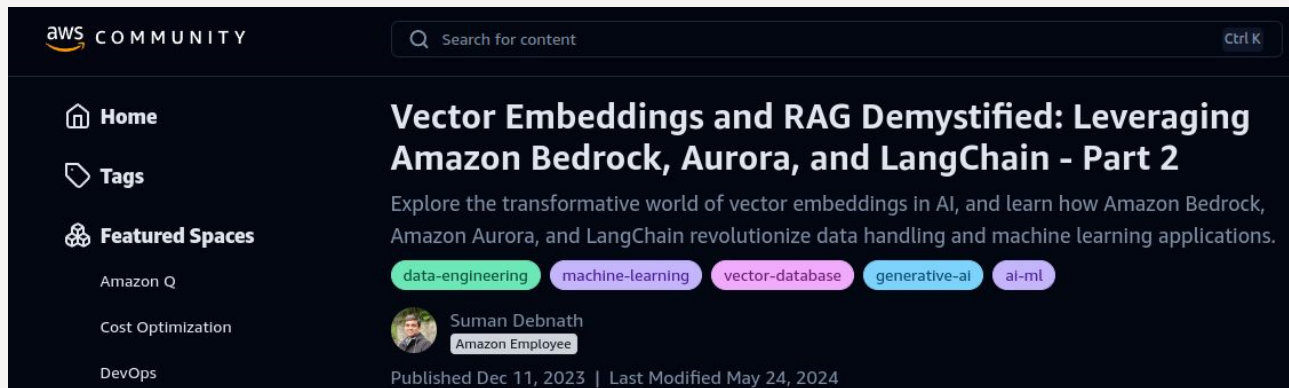
Chroma is the open-source AI application database. Batteries included.

Embeddings, vector search, document storage, full-text search, metadata filtering, and multi-modal. All in one place. Retrieval that just works. As it should be.

[ChromaDB Docs](#)



OTROS EJEMPLOS



The screenshot shows the AWS Community homepage. On the left, there's a sidebar with links to Home, Tags, and Featured Spaces (Amazon Q, Cost Optimization, DevOps). The main content area features a blog post titled "Vector Embeddings and RAG Demystified: Leveraging Amazon Bedrock, Aurora, and LangChain - Part 2" by Suman Debnath, an Amazon Employee. The post is tagged with "data-engineering", "machine-learning", "vector-database", "generative-ai", and "ai-ml". It was published on Dec 11, 2023, and last modified on May 24, 2024.



Amazon Kendra

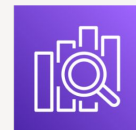
+



Amazon Bedrock



Amazon Aurora



OpenSearch



The screenshot shows the Azure Learn page for Azure Cosmos DB. The page has a dark theme with a navigation bar at the top. The main content area features a large heading "Generación aumentada de recuperación (RAG) en Azure Cosmos DB" and a subheading "En este artículo". The page also includes a sidebar with a search bar and a list of links to various Azure Cosmos DB resources.

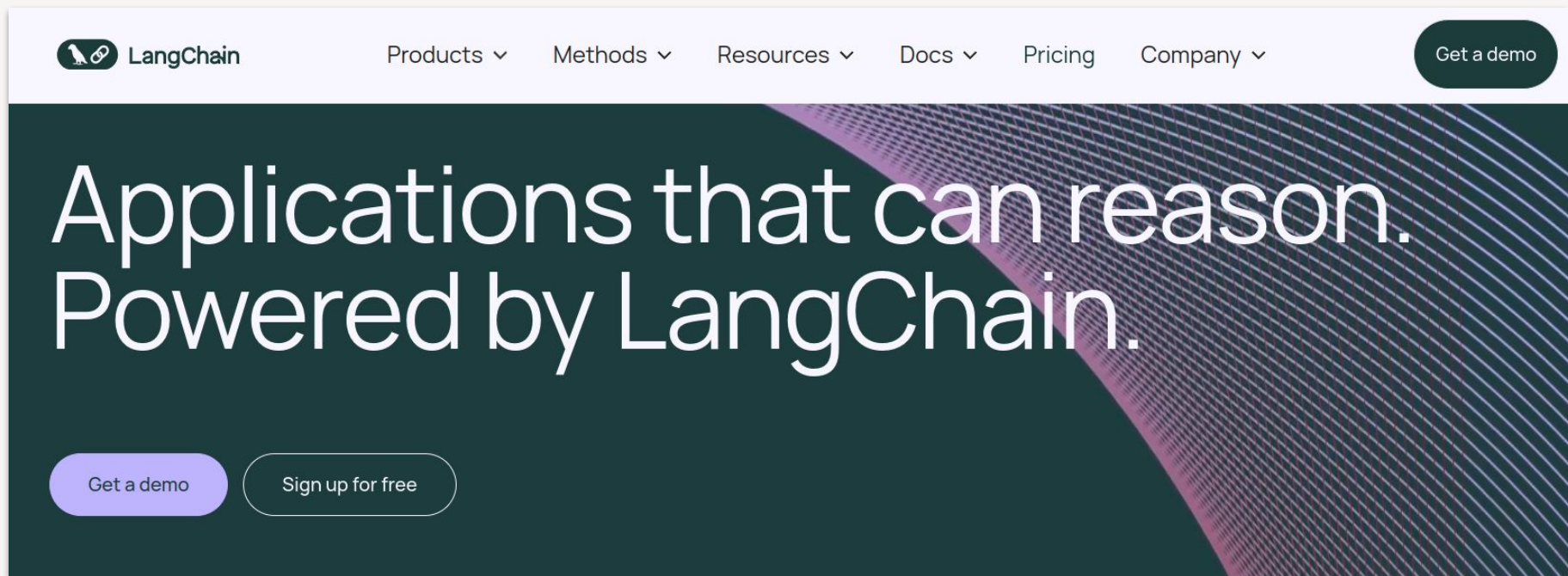
01
—

LANGCHAIN

Orchestration Framework

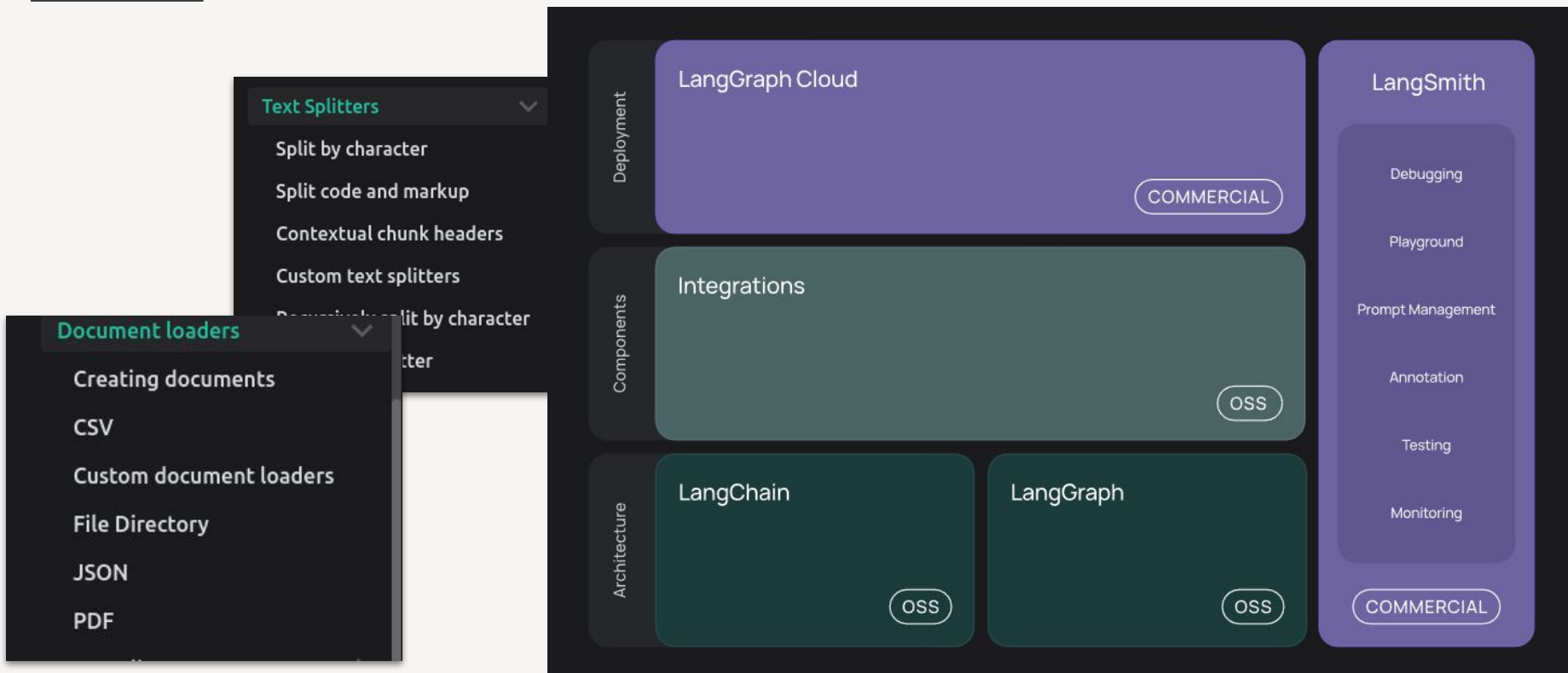
LANGCHAIN

Orchestration Framework para el desarrollo de aplicaciones con LLMs

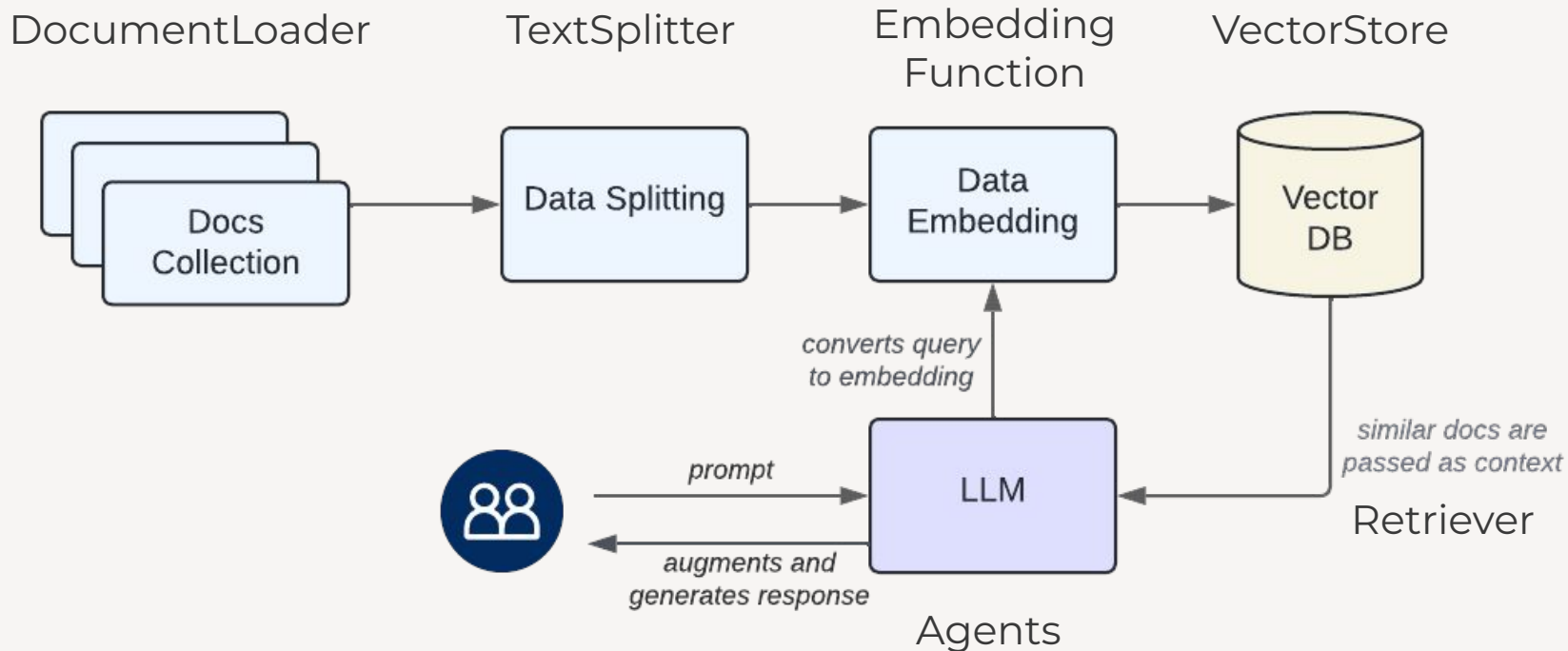


LANGCHAIN

[Langchain Doc](#)



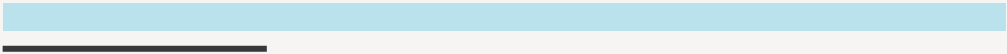
LANGCHAIN





NAIVE
RAG

¿Y por qué no aplicamos
finetuning en lugar de RAG?



RAG vs FT

	RAG	FT
Entorno	Dinámico, real-time	Más estático, requiere re-training
Caso de uso	Personalización y fácil actualización	Modificar comportamiento y estilo del modelo, RLHF
Recursos	“Menor” costo computacional	Alto costo computacional
Seguridad	Vulnerabilidades de Ing. software + LLM	Glitch Tokens, Jailbreak, Data Poisoning...

Se pueden combinar

RESUMEN

“Retrieve, augment and generate”

- RAG es un **modelo de arquitectura** para mejorar la eficacia de los LLMs
- Los componentes principales son el **indexing pipeline**, **generator** y **retriever**
- El indexing pipeline consiste en: doc **loading, splitting, embedding, store**
- RAG y **Fine-Tuning** son complementarios
- NO hay una única forma de implementar RAG



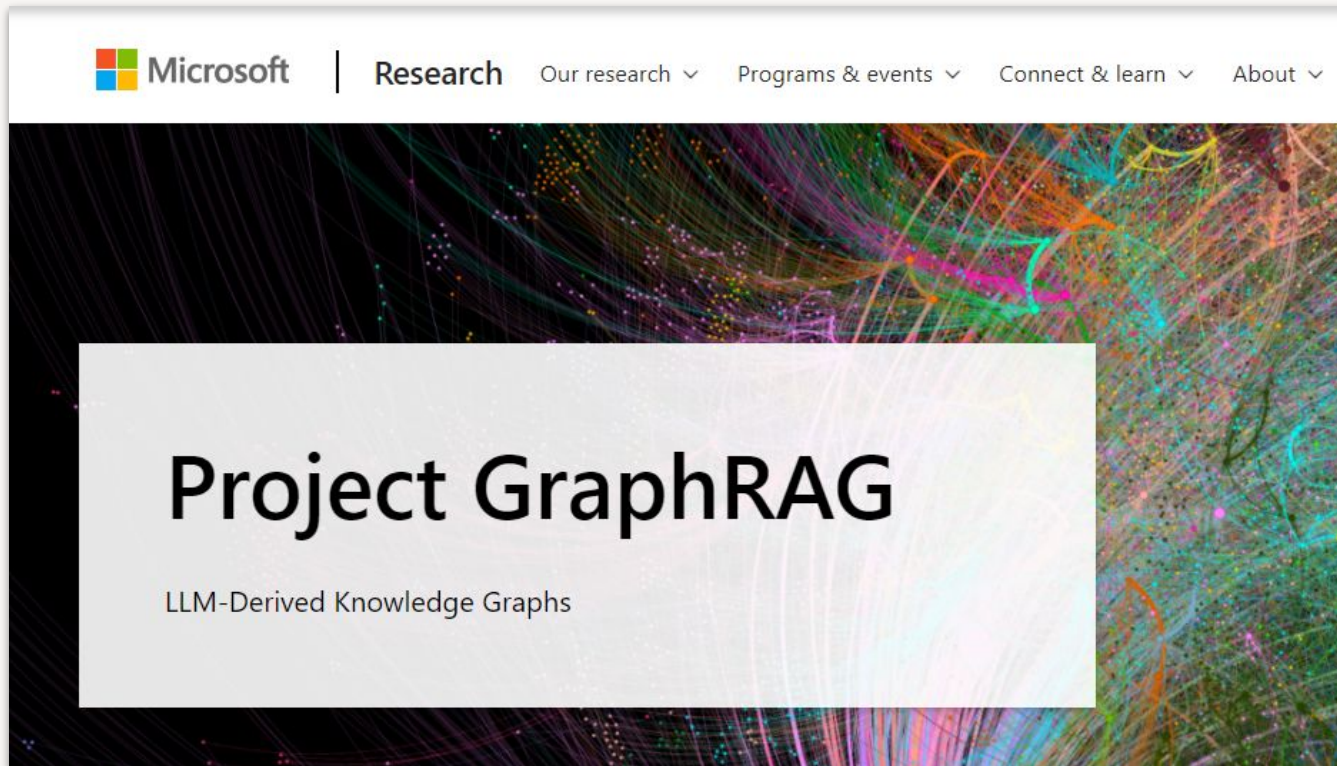
04.1

EXTRA

Curiosidades y noticias

CURIOSIDADES

[Microsoft GraphRAG](#)



Article | [Open access](#) | Published: 23 April 2024

Optimization of hepatological clinical guidelines interpretation by large language models: a retrieval augmented generation-based framework

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[npj Digital Medicine](#) **7**, Article number: 102 (2024) | [Cite this article](#)

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Abstract

Large language models (LLMs) can potentially transform healthcare, particularly in providing the right information to the right provider at the right time in the hospital workflow. This study investigates the integration of LLMs into healthcare, specifically focusing on improving clinical decision support systems (CDSSs) through accurate interpretation of medical guidelines for chronic Hepatitis C Virus infection management. Utilizing OpenAI's GPT-4 Turbo model, we developed a customized LLM framework that incorporates retrieval augmented generation (RAG) and prompt engineering. Our framework involved guideline conversion into the best-structured format that can be efficiently processed by LLMs to provide the most accurate output. An ablation study was conducted to evaluate the impact of different formatting and

Video y Paper sugeridos

Retrieval-Augmented Generation for Large Language Models: A Survey

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Abstract—Large Language Models (LLMs) showcase impressive capabilities but encounter challenges like hallucination, outdated knowledge, and non-transparent, untraceable reasoning processes. Retrieval-Augmented Generation (RAG) has emerged as a promising solution by incorporating knowledge from external databases. This enhances the accuracy and credibility of the

in Figure 1. The development trajectory of RAG in the era of large models exhibits several distinct stage characteristics. Initially, RAG's inception coincided with the rise of the Transformer architecture, focusing on enhancing language models by incorporating additional knowledge through Pre-Training Models (PTM). This early stage was characterized by work aimed at refining pre-training techniques. Subsequent arrival of ChatGPT [6] marked a turning point, with LLM demonstrating powerful in context capabilities. RAG research shifted towards providing information for LLMs to answer more complex, edge-intensive tasks during the inference stage, and development in RAG studies. As research on the enhancement of RAG was no longer limited to the pre-training stage but began to incorporate more with LLM inference techniques.

The emerging field of RAG has experienced swift growth,

Pregunta abierta

¿Qué atributos de seguridad se deberían considerar en RAG?

¿Qué vemos la clase próxima?



05

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¿Dónde puedo profundizar?

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¡ GRACIAS !